Matrix Theory

10 The four fundamental subspaces

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From description to action

- So far, we looked at the definitions: subspace, basis, etc.

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- We now compute explicit basis!

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- We now compute explicit basis!
- We can't construct by inspection, need systematic procedure.

Subspaces of a matrix we have seen already

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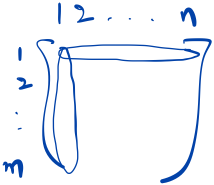
- Column space of A: $C(A)$
- Null space of A: $N(A)$

Column space $C(A)$

- $A_{m \times n}$

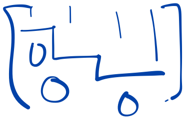
Column space $C(A)$

$\{b : Ax = b\}$



- $A_{m \times n}$
- Which vectors in $\mathbb{R}^m$ can be produced by A?

Column space $C(A)$



- $A_{m \times n}$
- Which vectors in $\mathbb{R}^m$ can be produced by A?
- Dimension is the rank r: no. of pivot columns

Null space $N(A)$

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- Which vectors in $\mathbb{R}^n$ are sent to zero?

Null space $N(A)$

$N(A) = \{ x : Ax = 0 \}$

- $A_{m \times n}$
- Which vectors in $\mathbb{R}^n$ are sent to zero?
- Dimension is $(n - r)$: no. of free variables

$1 \quad 2 \quad \cdots \quad n$

$x \in \mathbb{R}^n$

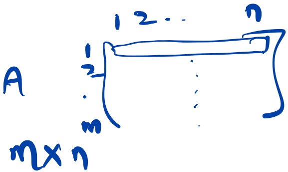
Let's consider A^T

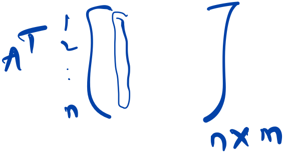
$$A^T \quad \underbrace{\begin{bmatrix} \; \end{bmatrix}}_{n \times m} \cdot \underbrace{\Big|}_{x \in \mathbb{R}^m} \longrightarrow \underbrace{\Big|}_{Ax \in \mathbb{R}^n}$$

- $A^T : \mathbb{R}^m \to \mathbb{R}^n$

Row space of A: $C(A^T)$

- Column space of A^T





$\boxed{\subseteq \mathbb{R}^n}$

Row space of A: $C(A^T)$

- Column space of A^T
- Spanned by the rows of A

Row space of A: $C(A^T)$

- Column space of A^T
- Spanned by the rows of A
- Dimension is the rank r: no. of independent rows

Left Null space of A: $N(A^T)$

- Null space of A^T

$Ax = 0$

Null space of A

free vars/cols

span $N(A)$

$\subseteq \mathbb{R}^n$

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Left Null space of A: $N(A^T)$

$N(A) \subseteq \mathbb{R}^n$

$Ax = 0$ $\quad m \times n$

$y^T \underset{m \times n}{A} = 0$

- Null space of A^T
- $\{y : A^T y = 0\}$

$m \times n$

$n - r$

Left Null space of A: $N(A^T)$

- Null space of A^T
- $\{y : A^T y = 0\}$
- Dimension is $(m - r)$: no. of rows beyond the independent ones

The four subspaces

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For $A \in \mathbb{R}^{m \times n}$:

Space	Meaning	Lives in	Dimension
$C(A)$	Column space	$\mathbb{R}^m$	r
$N(A)$	Null space	$\mathbb{R}^n$	$n - r$
$C(A^T)$	Row space	$\mathbb{R}^n$	r
$N(A^T)$	Left null space	$\mathbb{R}^m$	$m - r$

Example

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 2 & 4 & 6 \end{bmatrix}$$

3x3

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

R

$$C(A) = \text{span of} \left\{ \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} \right\}$$

$$N(A) = \text{span of} \left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

The column space: $C(A)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 2 & 4 & 6 \end{bmatrix} \rightarrow U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

The Null space of A: $N(A)$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 1 & 2 \\ 2 & 4 & 6 \end{bmatrix} \rightarrow U = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

The row space of A $C(A^T)$

- Nonzero rows of the echelon matrix form the basis

The row space of A $C(A^T)$

- Nonzero rows of the echelon matrix form the basis
- $C(A^T) =$ span of$\{(1, 2, 3), (0, -1, -1)\}$

Left Null space of A $N(A^T)$

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- Solve $A^T y = 0$

$$A^T = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 1 & 4 \\ 3 & 2 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$x_{sp} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$N(A^T) = \text{Span of}$ $\left\{ \text{1D subspace in inside } \mathbb{R}^3 \right\}$

Orthogonality of the subspaces

- Row space is orthogonal to Null space

$x \in N(A)$

$$A = \begin{bmatrix} - & r_1 & - \\ - & r_2 & - \\ & \vdots & \\ - & r_m & - \end{bmatrix} \begin{bmatrix} | \\ x \\ | \end{bmatrix}$$

$\{x : Ax = 0\}$

$$\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \begin{matrix} r_1^T x \\ r_2^T x \\ \vdots \\ r_m^T x \end{matrix}$$

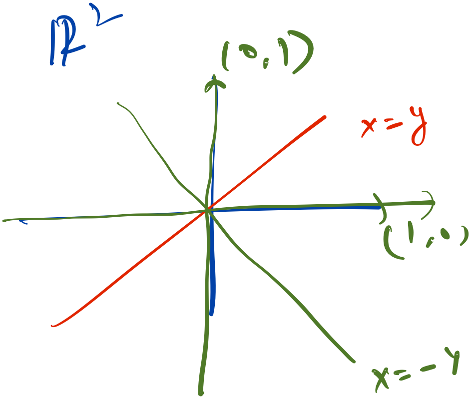
$r_i^T x = 0 \quad \forall i$

$(\sum c_i r_i)^T x = 0$

Orthogonality of the subspaces

- Row space is orthogonal to Null space
- If $x \in N(A)$, then $Ax = 0$

$\mathbb{R}^3$



Orthogonality of the subspaces

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Orthogonality of the subspaces

- Row space is orthogonal to Null space
- If $x \in N(A)$, then $Ax = 0$
- $N(A) = C(A^T)^{\perp}$, equivalently $C(A^T) = N(A)^{\perp}$

Column space is orthogonal to left null space

- If $y \in N(A^T)$, then $A^T y = 0 \rightarrow y$ is orthogonal to every column of A

rows of A^T $\perp$r to y

Cols of A $\perp$r to y

Column space is orthogonal to left null space

- If $y \in N(A^T)$, then $A^T y = 0 \rightarrow y$ is orthogonal to every column of A
- $N(A^T) = C(A)^{\perp}$

Four fundamental subspaces

$$
\begin{array}{ccc}
\mathbb{R}^n & & \mathbb{R}^m \\
C(A^T) & \xrightarrow{\;A\;} & C(A) \\
\dim = r & & \dim = r \\
\perp & & \perp \\
N(A) & & N(A^T) \\
\dim = n - r & & \dim = m - r
\end{array}
$$

$A_{m \times n}$

$x \in \mathbb{R}^n \xrightarrow{A_{m \times n}} Ax \in \mathbb{R}^m$

direct sum

$$\boxed{\mathbb{R}^n = C(A^T) \oplus N(A)}$$

$$\boxed{\mathbb{R}^m = C(A) \oplus N(A^T).}$$

$x = x_r + x_n$

every vector

In class exercise

$3 = C(A^T)$ $C(A) = 3$

$A_{m \times n}$

$5 = N(A)$ $N(A^T) = 2$ $\mathbb{R}^m$

$\mathbb{R}^n$

- For a 5×8 matrix of rank 3, what are the dimensions of all four subspaces?

In class exercise

$A_{m\times n}$ $\quad r = n$

- For a 5×8 matrix of rank 3, what are the dimensions of all four subspaces?
- If A has full column rank, what is $N(A)$?

$\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$

$r = n$

$A_{m\times n}$

$C_{n\times m} A_{m\times n} = I_{n\times n}$

In class exercise

- For a 5×8 matrix of rank 3, what are the dimensions of all four subspaces?
- If A has full column rank, what is $N(A)$?
- If A has full row rank, what is $N(A^T)$?

$m \times n$

In class exercise

- For a 5×8 matrix of rank 3, what are the dimensions of all four subspaces?
- If A has full column rank, what is $N(A)$?
- If A has full row rank, what is NA^T?
- If A is square and invertible, what are the four subspaces?

Another example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix} \xrightarrow{R} \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix}$$

$C(A) = \text{span of } \left\{ \begin{bmatrix} 1 \\ 3 \end{bmatrix} \right\}$ $\quad r = 1$

$N(A) = \text{span of } \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$ $\quad n - r = 1$

$C(A^T) = \text{span of } \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$ $\quad r = 1$

$\mathbb{R}^2 \neq \{ \text{span of } C(A^T) \cup \text{span of } N(A) \}$

$\mathbb{R}^2 = \text{span of } \{ C(A^T) \cup N(A) \}$ = direct sum of $C(A^T)$ & $N(A)$

The four subspaces